

TrES-3b

R-2487

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110527 · created 2026-08-24T00:56:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455708.92595 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.029
Transit depth (Rp/Rs)^2	3.85 +/- 1.13 [%]
Orbital Inclination (inc)	81.75 +/- 0.98
Airmass coefficient 1 (a1)	1.089 +/- 0.027
Airmass coefficient 2 (a2)	-0.081 +/- 0.083
Scatter in the residuals of the lightcurve fit is	2.51 %
Best Comparison Star	#2 - [553, 408]
Optimal Aperture	3.47
Optimal Annulus	14.31
Transit Duration (day)	0.057 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.029 vs 0.16309 (deviation 0.96σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.03σ)
Mid-time O–C	1.6 min
Depth SNR	5.7
Residual scatter	2.51 %

Light curve

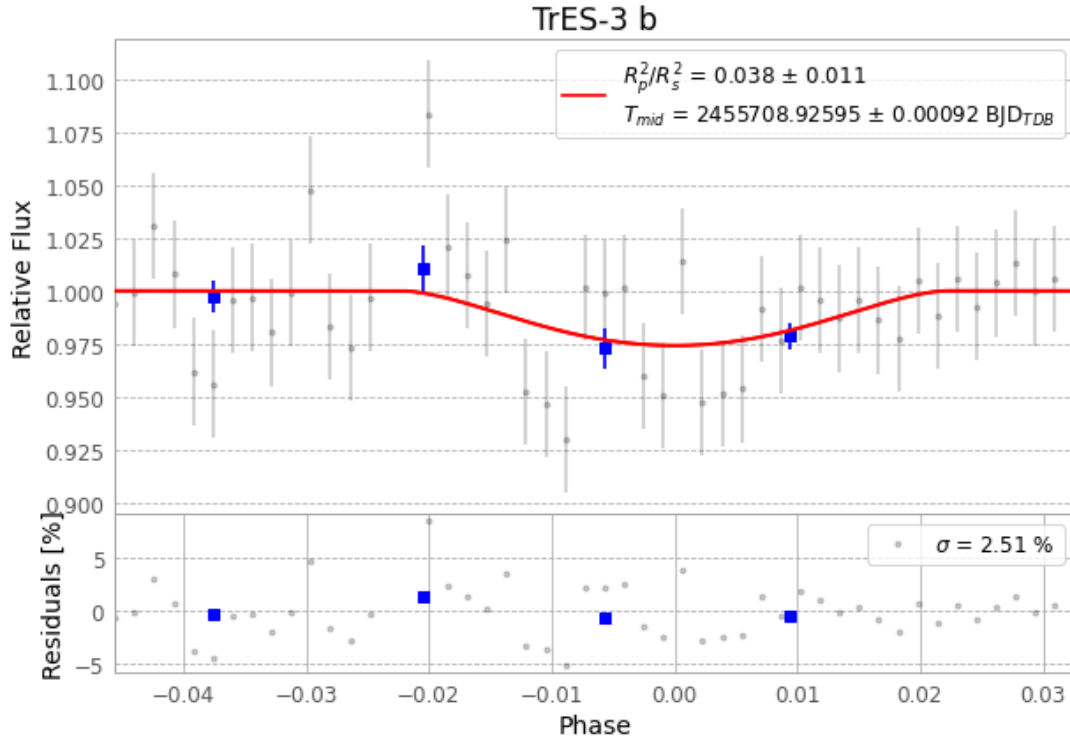


Figure 1 — FinalLightCurve_TrES-3 b_2011-05-27.png (SHA-256 643951a0f72b199e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 189], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 544251f65e2bf1ef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), TRES-3110527084213.FITS → TRES-3110527110913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110527.log (462e63272c83...), FinalLightCurve_TrES-3 b_2011-05-27.png (643951a0f72b...), FinalLightCurve_TrES-3 b_2011-05-27.csv (a5fea6109e5b...)

Compute resources

Wall time 497.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)